

NGUYEN MINH DUY

Data Engineer

Linh Xuan Ward, Thu Duc City, Ho Chi Minh City

☎ (+84) 795 455 127 ✉ minhduy200316@gmail.com 🌐 [linkedin.com/in/minzi03](https://www.linkedin.com/in/minzi03) 🐙 github.com/minzi03

Summary

Data Engineer with hands-on experience in **ETL/ELT**, **data modeling**, **batch and streaming processing**, and **Lakehouse architectures**. Focused on building **reliable, scalable, and analytics-ready data platforms**.

Education

VNU-HCM University of Information Technology (UIT)

Graduated: Jun 2026

Bachelor of Information Technology — Major in Information Systems

GPA: 3.1 / 4.0

- **Relevant Coursework:** Database Systems, Data Warehousing & OLAP, Big Data Analytics with Spark, Cloud Computing (Azure), Machine Learning & Data Mining, Business Intelligence.
- **English:** VSTEP B2 (CEFR B2).

Skills

Programming & Querying: Python (PySpark), SQL

Data Engineering & Modeling: ETL/ELT, Batch & Streaming, CDC, Star Schema, SCD Type 1/2, Medallion Architecture, Data Quality & Data Contracts

Big Data & Lakehouse: Apache Spark, Apache Iceberg, Delta Lake, MinIO (S3), Trino, Dremio

Streaming & Integration: Apache Kafka, Debezium, Spark Structured Streaming, Apache NiFi

Orchestration & Transformation: Apache Airflow, dbt, Azure Data Factory

Cloud & Warehouses: Azure (ADLS Gen2, Synapse Analytics), Databricks, Snowflake

BI & Governance: Power BI, Apache Superset, MS BI (SSIS/SSAS/SSRS), OpenMetadata

DevOps & Tools: Docker, Git, GitHub Actions (CI/CD)

Experience

Data Engineer Intern

May. 2026 – Present

QuanSkill (Remote, Part-time)

Python, PostgreSQL, Docker, GitHub Actions

- Built a **config-driven ingestion pipeline** for CSV, Excel, REST API, and PDF sources across **Raw → Staging → Clean** layers with validation, logging, file manifests, and PDF extraction.
- Defined **shared data contracts** for curated outputs consumed by **PostgreSQL/pgvector**, **RAG retrieval**, **document classification**, and **Streamlit reporting**.
- Containerized the application with **Docker Compose** and implemented **GitHub Actions CI/CD** for automated testing, staging validation, GHCR image publishing, and deployment/rollback.

Data Engineer Intern

Dec. 2025 – Mar. 2026

Katalyst (On-site)

Dremio, Iceberg, NiFi, MinIO, Oracle, SQL

- Designed **Apache Iceberg tables** and **Star Schema models** with **SCD Type 1 & 2** for analytical and reporting workloads on the Hanas Lakehouse Platform.
- Built **NiFi pipelines** from Oracle to MinIO and automated Dremio API workflows; tuned Raw and Aggregate Reflections to reduce query latency from **~25s to ~8–12s**.
- Implemented **Row Access Policies and Column Masking** and benchmarked **Dremio v25 vs. v24** across features, Reflections, and query performance to support platform upgrade evaluation.

Projects

Banking Data Platform | *Spark, Iceberg, Kafka, Debezium, Airflow, dbt, Trino, OpenMetadata, Superset, Docker* | *GitHub* **Present**

- Built a production-like **Medallion Lakehouse** (Bronze → Silver → Gold) with **Spark, Iceberg, and MinIO**; developed YAML-driven ETL for **16 source datasets**, producing **8 dimensions**, **5 fact tables**, and **4.6M+ curated financial transactions**.
- Implemented batch and near-real-time CDC pipelines with **PostgreSQL, Debezium, Kafka, and Spark Structured Streaming**; consolidated Bronze CDC events into Silver current-state tables using watermarks, deduplication, and **idempotent Iceberg MERGE**, achieving **sub-minute local freshness** across workflows orchestrated by **Airflow**.
- Delivered **Customer 360, RFM, churn-risk, and cross-sell analytics** through **Trino, dbt, and Superset**; added data contracts, quality checks, RBAC, column masking, and cataloged **53 tables with 22 lineage edges** in **OpenMetadata**.

Modern Data Stack Pipeline | *Kafka, Debezium, MinIO, Airflow, Snowflake, dbt, Docker* | *GitHub*

Dec. 2025

- Built a **CDC pipeline** (PostgreSQL → Debezium → Kafka → MinIO → Snowflake), using a Python micro-batch consumer to write **date-partitioned Parquet** with CDC metadata for customers, accounts, and transactions.
- Orchestrated idempotent ELT with **Airflow and dbt**, loading new files into Snowflake RAW and building incremental staging, dimension, and fact models with **SCD Type 2 snapshots**; automated validation and deployment with GitHub Actions.

Azure E-Commerce ETL Pipeline & Analytics | *ADF, ADLS Gen2, Databricks, Delta Lake, Synapse, Power BI* | [GitHub](#) **Jun. 2025**

- Built an end-to-end **Azure Lakehouse pipeline** (Bronze → Silver → Gold), ingesting MySQL, MongoDB, and HTTP API data into **ADLS Gen2** through parameterized **Azure Data Factory** pipelines.
- Developed **Databricks PySpark and Delta Lake** transformations with schema enforcement, OPTIMIZE, and Z-Ordering for a **2.9M-row fact table**; modeled curated data as a **Star Schema**, served it through Synapse and Power BI, and added automated failure alerts with Azure Monitor and Logic Apps.

Movie Data Warehouse (ETL & OLAP) | *SQL Server, SSIS, SSAS, SSRS, Power BI* | [GitHub](#) **Jul. 2024**

- Designed a **Star Schema** with a **~780K-row fact table and 8 dimensions** from 1M+ raw records, and built **SSIS ETL pipelines** for data cleansing, transformation, and deduplication.
- Developed **SSAS OLAP cubes** and MDX queries, and delivered **SSRS/Power BI reports** for box office revenue, studio performance, and other analytical KPIs.

Awards & Certifications

- **IBM (2026):** Data Engineering Professional Certificate.
- **Microsoft (2025):** DP-203 Data Engineering on Azure, DP-900 Azure Data Fundamentals, DP-3027 Databricks.
- **DataCamp (2025):** Data Engineer; Data Engineer Associate.
- **Axon Active Vietnam (2025):** Software Development with Scrum.
- **Google (2024):** Professional Certificates — Data Analytics, Advanced Data Analytics, Business Intelligence.